

CADET: Physics-Grounded Causal Auditing of End-to-End Driving Planners

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Abstract—End-to-end (E2E) autonomous-driving planners trained by imitation can exploit spurious correlations: they link scene elements that merely co-occur with expert actions, such as a roadside object or a building facade, to driving decisions, instead of the variables that causally determine them. Such causal confusion is hard to detect, because the prevailing open-loop metrics, L2 displacement and collision rate, are dominated by ego status and say little about whether a planner depends on spurious cues. Existing remedies based on causal-intervention training require retraining and cannot audit an already deployed planner. We present CADET, a training-free framework that measures spurious reliance in a pretrained E2E planner and localizes it to individual planning queries, with no parameter update. CADET has three components. First, a Physics-grounded Causal Reliance score (PCR) identifies planning queries that the model depends on but that cannot physically influence the decision. It fuses perturbation-based model influence with a physics-geometric prior (time-to-collision, drivable-corridor relevance, and object class). The prior is read off scene geometry, so it is an external anchor that is independent of the training distribution. Second, a counterfactual robustness benchmark with three perturbation families (spurious, causal-link, and distribution-shift) and three metrics (CSI, CRI, and CCS). We show that the natural stability probe is degenerate: it is satisfied by construction whenever it selects its targets with the same score the repair masks by. We give two probes that are decoupled from it. Third, Test-time Causal Masking (TCM) suppresses the flagged queries at inference, which shows that the located dependencies are removable without training. On a controlled benchmark in which the status of every object is fixed by construction, PCR localizes spurious reliance at an F1 of 0.92 against 0.50 for the next-best training-free criterion, and it alone reaches that precision while masking under 13% of queries and keeping causal response intact. On the pretrained SparseDrive planner evaluated on nuScenes, the audit measures a causal-to-spurious per-agent influence ratio of 9.5, yet in individual frames the most influential physically irrelevant agent displaces the plan as strongly as a genuine causal agent. Suppressing those agents moves the official open-loop metrics by under 0.1 m, which shows that displacement error does not register a change in causal reliance and is therefore unsuited to measuring it.

Index Terms—End-to-end autonomous driving, causal confusion, spurious correlation, causal intervention, counterfactual robustness, trustworthy AI, training-free.

I. INTRODUCTION

END-TO-END (E2E) autonomous driving has become the dominant paradigm since UniAD [1], with VAD [2], SparseDrive [3], and PARA-Drive [4] further improving planning accuracy and efficiency. These systems are trained by

imitation on large logged datasets, which introduces a structural weakness: imitation learning is non-causal and can exploit spurious correlations, namely features that co-occur with expert actions in the data but do not cause them [5]. A planner may, for instance, learn to decelerate near a particular building facade or roadside object because such decelerations were frequent in the logs, while disregarding the agent that genuinely warrants braking. This causal confusion is a principal obstacle to advancing E2E driving from L2 to L3/L4 reliability.

Two factors make this failure mode hard to address. First, it is hard to observe. Evaluation still relies mainly on open-loop L2 displacement and collision rate on nuScenes. These metrics are dominated by ego status and change little when the perception input is removed [6], so a low L2 error says little about whether a planner relies on spurious cues. Second, it is costly to repair. Recent causal-intervention methods such as CausalVAD [7] reduce confounding by retraining the planner with a sparse intervention module. Retraining is expensive, and it gives no way to audit a planner that has already been trained and deployed. What is missing is a training-free tool that determines whether, where, and how much a given pretrained planner relies on spurious correlations.

We address this gap with CADET (Causal Auditing and Deconfounding at Test-time), a training-free framework whose primary object is measurement: given a frozen planner, it determines which of the planner’s own queries carry reliance that physics cannot justify, and it does so without parameter updates (Fig. 1). Our central observation is that true causes cannot be distinguished from globally spurious correlates on the basis of training-data statistics alone: when a spurious co-occurrence holds across all environments, observational signals such as perturbation sensitivity and cross-environment invariance both fail under the same faithfulness assumption [8]. We therefore introduce an external anchor: a physics-geometric prior derived from the perception module’s own outputs (3D boxes, velocity, class, and detection confidence). This prior determines whether an object is capable of influencing the decision at all, through time-to-collision, drivable-corridor relevance, and dynamic or static state, independently of the data distribution. A query on which the planner relies heavily, yet which the physics prior deems irrelevant, is by construction a spurious dependency.

Contributions.

- **Physics-grounded causal audit (PCR).** A training-free, model-agnostic score that flags per-query spurious re-

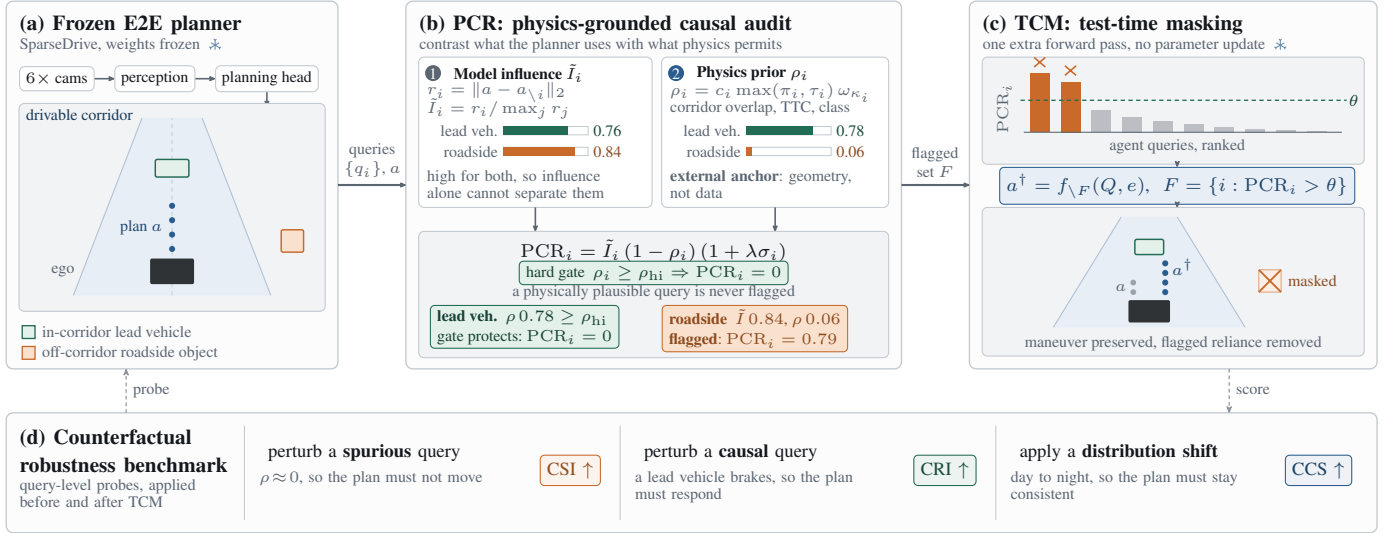


Fig. 1: Overview of CADET. (a) A pretrained end-to-end planner is run unmodified and emits vectorized agent queries together with an ego plan. (b) PCR audits each query by contrasting two signals: the model influence $\tilde{I}_i$, obtained by ablating the query and measuring the induced plan change, and a physics-geometric prior ρ_i computed from drivable-corridor overlap, time-to-collision, and object class. The planner relies on the in-corridor lead vehicle and on the off-corridor roadside object to a comparable degree, so influence alone cannot separate them. Because ρ is read off scene geometry rather than training statistics, it acts as an external anchor, and a hard gate on ρ guarantees that a physically plausible query is never flagged. The values shown are the per-category means measured on SpurGen (Table I). (c) TCM suppresses the flagged set at inference and re-plans, which removes the spurious reliance while preserving the maneuver. (d) A counterfactual benchmark probes the same frozen planner along three perturbation families and scores it with CSI, CRI, and CCS. The pipeline requires no parameter update and runs inference-only on a single 16 GB GPU.

liance by fusing perturbation-based model influence with a physics-geometric prior, which acts as a hard external arbiter, together with a cross-environment stability term. It requires only forward passes on a pretrained planner (Section IV).

- **Counterfactual robustness benchmark, with a probe design that is not self-fulfilling.** A standardized protocol with three perturbation families (spurious, causal-link, and distribution-shift) and three metrics, the Causal Stability Index (CSI), Causal Response Index (CRI), and Causal Consistency Score (CCS). We show that the obvious way to build the stability probe is degenerate, since selecting its targets by the physics score that the repair also masks by makes the probe a null operation, and we replace it with a ground-truth probe and an insertion probe that no masking rule can satisfy by deletion (Section IV-B, Section VI-B).
- **A demonstration that the located reliance is removable (TCM).** Test-time Causal Masking suppresses the flagged queries at inference, realizing an approximate $\text{do}(\cdot)$ intervention at deployment with no training. TCM shows that the audit localizes an actionable set: the flagged dependencies can be removed at test time while the maneuver is preserved (Section VI-E).
- **Reproducibility on commodity hardware.** The full pipeline is inference-only and runs on a single 16 GB GPU. We release the code, the benchmark, and the audit

toolkit.

CADET is complementary to existing causal methods. Unlike CausalVAD [7] and Beyond Patterns [9], which re-train models for planning or prediction, and unlike generative world-model approaches, which synthesize counterfactual scenarios for closed-loop testing, CADET neither trains nor generates: it instruments existing planners and is therefore directly applicable to deployed systems. The remainder of the paper reviews related work (Section II), formalizes the causal setting (Section III), details CADET (Section IV), describes the protocol (Section V), reports the audit (Section VI), and discusses implications and scope (Section VII, Section VIII).

II. RELATED WORK

A. End-to-End Driving Planners

Deep learning now covers tasks as varied as efficient transformer architectures [10], model compression for question answering [11], generative modeling [12], speech enhancement [13], visual recognition [14], reinforcement learning for combinatorial optimization [15], and multi-robot coordination and scheduling [16], [17]. Within this progress, vision-centric driving models have grown into end-to-end (E2E) systems that map raw sensor input directly to a plan. UniAD [1] introduced a planning-oriented architecture that unifies perception, prediction, and planning; VAD [2] replaced dense rasterization with a vectorized scene representation; and SparseDrive [3] and PARA-Drive [4] improved efficiency through sparse queries

and parallelized auxiliary tasks. These planners are trained by imitation and are evaluated mainly by open-loop L2 displacement and collision rate on nuScenes. Li et al. [6] show that these metrics are dominated by ego status: removing the perception input leaves the reported values almost unchanged. A low open-loop error therefore says little about whether a planner reasons over the scene or exploits spurious regularities. This is what the measure of causal reliance developed here is for.

B. Causal Confusion and Deconfounding in Driving

The non-causal nature of behavioral cloning was established by de Haan et al. [5], who showed that imitators rely on effects rather than causes, such as a brake-indicator light, so that additional observation can yield worse policies under distribution shift. In driving, CausalVAD [7] is the closest precursor to our work: it instantiates do-calculus as a sparse causal intervention scheme, builds a dictionary of context prototypes, and performs backdoor adjustment on vectorized queries, achieving state-of-the-art open-loop planning. Beyond Patterns [9] applies a diffusion-based backdoor adjustment to map features for trajectory prediction. Closer to the imitation-planning literature, PLUTO [18] and PlanTF [19] curb spurious reliance with training-time data augmentation, dropping the leading agent or masking the ego state so the planner cannot lean on them. All of these methods retrain the planner, and each identifies confounders from learned features or predefined attributes, without an external, distribution-independent criterion for causal status. CADET differs on both counts: it operates on a frozen planner without parameter updates, and it anchors the spurious-versus-causal distinction in a physics-geometric prior rather than in data statistics. This covers the global-spurious-correlation case in which feature-based identification fails. We position CADET against the families of work relevant to deployed systems: causal-intervention deconfounders (CausalVAD [7], Beyond Patterns [9], PLUTO [18], PlanTF [19]), the behavioral-cloning analyses that first exposed causal confusion [5], invariance learning [8], [20], perturbation attribution [21], and generative or closed-loop evaluation [22], [23]. Across these families, CADET is the only approach that operates on a frozen planner, repairs it at inference, and grounds the causal decision in an external, distribution-independent physics anchor. The training-free modules that are applicable to a frozen planner are compared quantitatively in Table IV. The retraining-based and other test-time methods are positioned here instead, because they cannot be applied to a frozen model as a flagging module. End-to-end planners such as UniAD, VAD, and SparseDrive are not competitors here but the subjects of the audit: CADET instruments a given planner, and its interface is planner-agnostic (Section V-B).

C. Counterfactual Evaluation and Robustness

A complementary line of work evaluates robustness through interventions. Generative world models synthesize safety-critical counterfactual scenarios for closed-loop testing, and Bench2Drive [22] and NAVSIM [23] provide closed-loop and

non-reactive simulation benchmarks. Both are motivated by the gap between open-loop L2 and closed-loop behavior. Invariance-based methods such as invariant risk minimization [8] seek predictors that are stable across environments, but they rest on a faithfulness assumption that fails precisely when a spurious correlate is invariant across all observed environments. Our counterfactual benchmark targets a different goal, namely measuring how much a given pretrained planner relies on spurious versus causal factors. It uses inexpensive, deterministic query-level perturbations rather than a generative simulator, so it is reproducible on commodity hardware.

a) *Plug-and-play, test-time methods.*: Because CADET operates on a frozen planner at inference, its true peers are plug-and-play, training-free methods applied at test time. Recent work in this vein improves a deployed planner without retraining: TOAD [24] optimizes the output trajectory at test time with a sampling search, and Centaur [25] adapts the planner online through test-time training. These methods target overall driving quality rather than per-agent spurious reliance. TOAD’s trajectory search and Centaur’s online updates are orthogonal to, and composable with, our masking operator. The criteria directly comparable to CADET are those that, like it, flag or suppress agent queries at inference: perturbation attribution (occlusion analysis) [21] ranks queries by the output change they cause; a training-free invariance test in the spirit of ICP/IRM [8], [20] flags queries whose influence is unstable across environments; and simple detector-driven heuristics (masking low-confidence or randomly selected queries) require no signal at all. All of these criteria can drive the same test-time masking operator, and we therefore benchmark against them across several planners in Table IV. We find that the influence- and invariance-based criteria improve stability only by masking genuinely causal agents as well. CADET differs from this family through the external physics anchor, which is what lets it suppress spurious reliance while keeping causal response.

D. Scene Perception and Structural Priors

The physics prior in CADET is computed from the perception module’s own outputs, so understanding street scenes from imagery is an enabling capability, recently advanced by combining large language models with street-view data [26]. Using a structural or physical prior to guide decision-making follows the same idea as its use in balanced and scalable multi-robot path planning [27]. CADET applies the same principle to E2E driving, using a kinematic prior as an external anchor against spurious reliance.

III. PROBLEM FORMULATION

A. A Structural Causal Model for Planning

We model an end-to-end planner with a structural causal model (SCM) $\mathcal{M} = (\mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}))$. The endogenous variables $\mathbf{V}$ comprise a latent *scene context* C (weather, road geometry, traffic density, and data-collection biases), the *perception state* S produced by the perception stack, and the *plan* Y (the future ego trajectory). The exogenous $\mathbf{U}$ collect

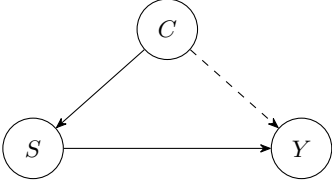


Fig. 2: Structural causal model of end-to-end planning. Scene context C confounds perception state S and plan Y via the backdoor path $Y \leftarrow C \rightarrow S$ (dashed). An imitation-trained planner fits the confounded $P(Y | S)$, whereas CADET targets the deconfounded $P(Y | \text{do}(S))$.

unobserved environmental noise. The relevant causal structure (Fig. 2) is

$$C \rightarrow S, \quad C \rightarrow Y, \quad S \rightarrow Y, \quad (1)$$

so that C is a *confounder* opening a backdoor path $Y \leftarrow C \rightarrow S$. An imitation-trained planner fits the observational $P(Y | S)$, which is biased by this backdoor path. The quantity of interest is instead the interventional $P(Y | \text{do}(S))$. Under the backdoor criterion with adjustment set C ,

$$P(Y | \text{do}(S)) = \sum_c P(Y | S, C=c) P(C=c). \quad (2)$$

Eq. (2) is the objective that training-time methods such as CausalVAD [7] approximate through retraining. Our objective differs: we audit and approximate this adjustment at test time on a frozen planner. The tendency of modern models to exploit context-induced spurious cues rather than causal ones is not specific to driving. It recurs across modalities, for instance as sycophancy in video and medical vision–language models [28], [29], which motivates a general, model-agnostic treatment.

B. Query-Level Notation

Modern planners [1]–[3] expose a set of vectorized queries $Q = \{q_1, \dots, q_N\}$, each q_i encoding a perceived entity (agent or map element) together with physical attributes $\phi_i = (\mathbf{b}_i, \mathbf{v}_i, \kappa_i, c_i)$: 3D box, velocity, class, and detection confidence. With ego state e , the planner emits a trajectory $a = f(Q, e) \in \mathbb{R}^{T \times 2}$ over a horizon T . We define a *removal operator* m_i that replaces q_i by a baseline (mean query or null token),

$$f_{\setminus i}(Q, e) \triangleq f(m_i(Q), e), \quad (3)$$

and a set version $f_{\setminus F}$ for a subset $F \subseteq \{1, \dots, N\}$.

C. Spurious vs. Causal Reliance

The *reliance* of the plan on query i is the decision change under its removal,

$$r_i = \|f(Q, e) - f_{\setminus i}(Q, e)\|_2. \quad (4)$$

A query is *causally relevant* if the entity it represents can physically affect the ego decision, such as a lead vehicle or a crossing pedestrian. It is *spurious* if r_i is large yet the entity cannot physically influence that decision, as in the case of

a distant static object off the ego path. The central difficulty is that observational signals alone cannot separate the two: when a spurious co-occurrence holds across all training environments (a global spurious correlation), both a high reliance r_i and a low cross-environment variance of r_i wrongly indicate causality. This is the failure of the faithfulness assumption that limits invariance-based methods [8]. We therefore introduce an external, distribution-independent anchor, namely a physics-geometric prior, in Section IV.

IV. THE CADET FRAMEWORK

CADET comprises three training-free components: a per-query audit score (Section IV-A), a counterfactual benchmark (Section IV-B), and a test-time fix (Section IV-C). All operate on a frozen planner using forward passes only. An overview is shown in Fig. 1, and Fig. 3 walks the audit through a single real nuScenes frame.

A. Physics-Grounded Causal Reliance Score (PCR)

For each query we combine three complementary signals.

a) (a) *Model influence.*: The (normalized) reliance from Eq. (4),

$$\tilde{I}_i = r_i / \max_j r_j \in [0, 1], \quad (5)$$

captures the planner’s current dependence structure. It needs no supervision, but it reflects learned behavior rather than ground-truth causality, so it is only a first-stage filter.

b) (b) *Physics-geometric prior (hard arbiter).*: From the perception attributes ϕ_i we compute a distribution-independent plausibility $\rho_i \in [0, 1]$ that the entity can influence the ego decision:

$$\rho_i = c_i \cdot \max(\pi_i, \tau_i) \cdot \omega_{\kappa_i}, \quad (6)$$

where c_i is detection confidence; $\pi_i \in [0, 1]$ is *path relevance*, the overlap of the entity (and its short-horizon motion forecast from $\mathbf{v}_i$) with the ego drivable corridor; $\tau_i = \text{clip}(1 - \text{TTC}_i / T_{\max}, 0, 1)$ is a time-to-collision urgency that vanishes for static, off-path, or diverging entities ($\text{TTC}_i \rightarrow \infty$); and ω_{κ_i} is a class weight (e.g., vulnerable road users upweighted). Because ρ_i is computed from physical kinematics and scene geometry rather than from training-data statistics, it is invariant to global spurious correlation, and it is therefore the external anchor.

c) (c) *Cross-environment stability.*: Partitioning scenes into environments $\mathcal{E}$ (weather/road/time), we measure the dispersion of $\tilde{I}_i$ across $\mathcal{E}$, $\sigma_i \in [0, 1]$. High dispersion indicates locally spurious reliance that signal (b) may fail to detect.

d) *Fusion.*: The reliance score is high only when the planner relies on a query that physics deems implausible:

$$\text{PCR}_i = \tilde{I}_i \cdot (1 - \rho_i) \cdot (1 + \lambda \sigma_i), \quad (7)$$

with the physics prior applied as a *hard gate*:

$$\text{PCR}_i \leftarrow \begin{cases} 0, & \rho_i \geq \rho_{\text{hi}} \text{ (protect causal queries),} \\ \text{PCR}_i, & \rho_i \leq \rho_{\text{lo}}, \\ \text{soft (7),} & \text{otherwise.} \end{cases} \quad (8)$$

The gate guarantees that a physically plausible query is never flagged. The policy is conservative: a query is suppressed only when the physics prior confirms it to be implausible.

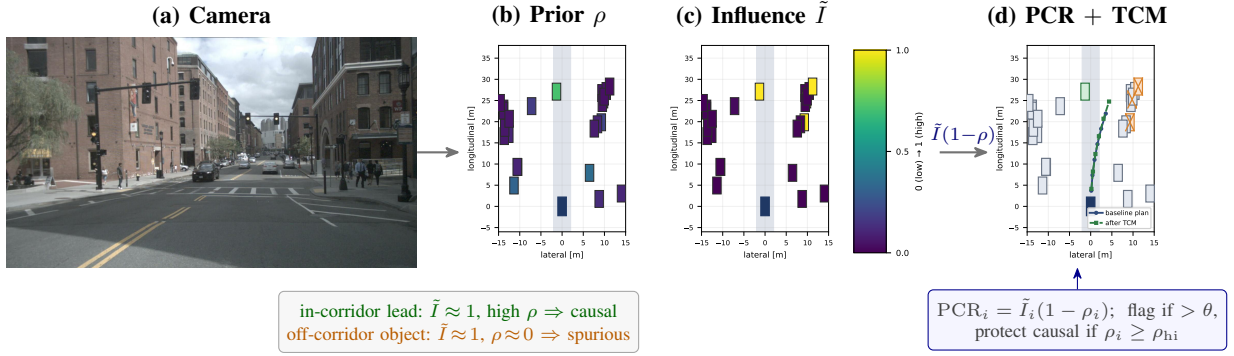


Fig. 3: CADET on a real nuScenes frame (audit of frozen SparseDrive). **(a)** The front camera view. **(b)** The physics prior ρ per detected agent (brighter = more able to affect the ego): the in-corridor lead vehicle scores high, off-corridor roadside objects $\rho \approx 0$. **(c)** The model influence $\tilde{I}$ from query ablation: the planner relies heavily ($\tilde{I} \approx 1$) not only on the lead vehicle but also on several off-corridor objects. **(d)** PCR = $\tilde{I}(1 - \rho)$ with the hard gate isolates the off-corridor objects as spurious (orange, crossed) while protecting the physically plausible lead (green); masking the flagged agents (TCM) yields the deconfounded plan (dashed), which proceeds without the spurious caution. An influence-only criterion would instead flag the genuine lead vehicle, because it cannot tell high reliance from physical relevance.

B. Counterfactual Robustness Benchmark

We perturb scenes at the query/perception level (cheap, training-free, deterministic) along three families and define a metric for each. Let $d(a, a') = \|a - a'\|_2$ be a normalized plan distance.

a) Spurious perturbation $\rightarrow$ CSI: A stability metric must not select its probe with the same quantity that the repair uses to decide what to suppress. Deleting the queries with $\rho_i \approx 0$ is the natural first formulation, but a module that masks by ρ has already removed them, so the probe becomes a null operation and the score approaches 1 whether or not the audit was correct. The degeneracy is not hypothetical: a mask that suppresses every query with $\rho_i < \rho_{lo}$ attains a score of exactly 1.00 on every planner we test (Table III). We therefore define the probe by criteria that do not consult ρ .

Ground-truth probe (CSI-gt). In a controlled benchmark the status of each object is fixed by construction, so we delete the objects that are unambiguously unable to affect the decision, those that are static and laterally clear of the corridor by a margin. This property is read off the generative parameters and uses neither time-to-collision, nor class weight, nor detection confidence, and therefore shares no factor with ρ . Objects inside the margin are treated as ambiguous and left in place, so a planner is never charged for responding to an object that could in fact matter. Writing a^{gt} for the resulting plan,

$$\text{CSI-gt} = 1 - \mathbb{E}[d(a, a^{gt})] \quad (\uparrow \text{ better}). \quad (9)$$

Insertion probe (CSI-ins). We insert one further irrelevant object at a lateral offset sampled clear of the corridor and compare the plans that the complete pipeline, audit and masking included, produces with and without it,

$$\text{CSI-ins} = 1 - \mathbb{E}[d(a^\dagger, a_+^\dagger)] \quad (\uparrow \text{ better}), \quad (10)$$

where $a_+^\dagger$ is the plan on the augmented scene. The inserted object did not exist when any audit was computed, so no flagging rule can have been fitted to it, and the metric measures whether the rule generalizes. Nor can the score be raised by

masking more of the original scene, since the reference plan $a^\dagger$ moves under the same intervention. We report CSI-ins as the evidence of deconfounding and retain the ρ -defined variant, written CSI- ρ , only to quantify how far it overstates the effect.

b) Causal-link perturbation $\rightarrow$ CRI: Modify a causal query (ρ_i high, e.g., lead-vehicle braking). The plan should respond. With a^{ca} and the expected correct response direction Δ^* ,

$$\text{CRI} = \mathbb{E}[\mathbb{1}\{\langle a^{ca} - a, \Delta^* \rangle > 0\}] \quad (\uparrow \text{ better}). \quad (11)$$

c) Distribution-shift perturbation $\rightarrow$ CCS: Apply a style shift preserving causal structure (e.g., day $\rightarrow$ night), so the plan should stay consistent,

$$\text{CCS} = 1 - \mathbb{E}[d(a, a^{ds})] \quad (\uparrow \text{ better}). \quad (12)$$

Unlike generative counterfactual world models, these perturbations are query-level edits, so the benchmark is reproducible on commodity hardware.

C. Test-time Causal Masking (TCM)

At inference we suppress the flagged spurious set $F = \{i : \text{PCR}_i > \theta\}$ and recompute the plan,

$$a^\dagger = f_{\setminus F}(Q, e), \quad (13)$$

an approximate $\text{do}(\cdot)$ that removes spurious confounding at deployment time. Optionally, rather than nulling q_i , we average over a context prototype dictionary $\{p_c\}$ (a test-time analogue of the backdoor adjustment in Eq. (2)), $a^\dagger = \frac{1}{|C|} \sum_c f(Q \mid q_i \leftarrow p_c, e)$. By the hard gate in Eq. (8), F never contains a physically plausible query; consequently TCM cannot remove a true cause and preserves clean-scene accuracy by construction. TCM incurs a single (batched) forward pass and requires no parameter update. Algorithm 1 summarizes the pipeline.

Algorithm 1 CADET audit and test-time deconfounding

Input: frozen planner f , queries Q , ego e , attributes $\{\phi_i\}$, threshold θ

for each query i **do**

$r_i \leftarrow \|f(Q, e) - f_{\setminus i}(Q, e)\|_2$ // influence

$\rho_i \leftarrow c_i \cdot \max(\pi_i, \tau_i) \cdot \omega_{\kappa_i}$ // physics prior

end for

$\tilde{I}_i \leftarrow r_i / \max_j r_j$; compute σ_i across environments

$\text{PCR}_i \leftarrow \tilde{I}_i(1 - \rho_i)(1 + \lambda\sigma_i)$, apply hard gate (8)

$F \leftarrow \{i : \text{PCR}_i > \theta\}$

$a^\dagger \leftarrow f_F(Q, e)$ // test-time causal masking

return audit $\{\text{PCR}_i\}$, deconfounded plan $a^\dagger$

V. EXPERIMENTAL SETUP

We evaluate CADET in two complementary settings: a controlled synthetic study in which the spurious or causal status of every agent is known by construction (Section V-A), and an audit of a public pretrained planner on real nuScenes data (Section V-B). Both are inference-only and run on a single NVIDIA RTX 2000 Ada GPU (16 GB).

A. SpurGen: a Physics-Grounded Controlled Benchmark

Real driving data offer no ground truth for which dependencies are spurious, so we construct SpurGen, a controlled benchmark in which the causal status of every object is known by design. Each of its 400 scenes is generated in one of three environments (sunny, rain, night) that modulate detection confidence and clutter density, and contains: an in-path closing agent; a cut-in vehicle approaching the lane laterally; one to five off-path static distractors (“trees”); in 40% of scenes an off-path “billboard”; and, in half of the scenes, a static “mailbox” that is usually off-path but occasionally near the lane edge (1781 object queries in total, 255 spurious-reliance instances). A mock planner brakes for the causal hazards (in-path closing agents with low time-to-collision, cut-in vehicles entering the corridor), with two realistic environment-dependent causal behaviors (extra caution for pedestrians at night and for cut-in vehicles in rain), and carries two deliberately injected shortcuts that span the two spurious regimes of Section III-C: a global one (it brakes in the presence of a mailbox, with a per-object strength, in every environment) and a local one (it brakes for billboards only in sunny scenes, a glare-like association). Both shortcut objects are spurious by construction, so the benchmark has a label-free gold standard. The cut-in vehicle and the edge-of-lane mailbox provide realistic hard cases. Noisy variants add Gaussian perception noise (standard deviation σ up to 1.5) to positions, velocities, and confidence. The implementation depends only on NumPy, and the generator is released with the benchmark.

Table I summarizes the composition, and Fig. 4 shows why the benchmark is diagnostic. Spatially (Fig. 4(a)), causal agents concentrate in and around the ego corridor while distractors and mailboxes lie outside it. In the influence–physics plane (Fig. 4(b)), the three roles separate: causal agents have both high influence and high ρ (protected region), benign

TABLE I: SpurGen composition (400 scenes; 134/133/133 across sunny/rain/night). Mean normalized influence $\tilde{I}$ and physics prior ρ per category: the spurious objects combine near-zero ρ with non-trivial influence, the signature CADET is designed to detect.

Category	Count	Role	Mean ρ	Mean $\tilde{I}$
In-path / cut-in vehicle	402	causal	0.78	0.76
Pedestrian	153	causal	0.82	0.74
Static distractor (tree)	851	benign	0.01	0.01
Mailbox (global shortcut)	204	spurious	0.06	0.84
Billboard (sunny-only shortcut)	171	spurious	0.00	0.25

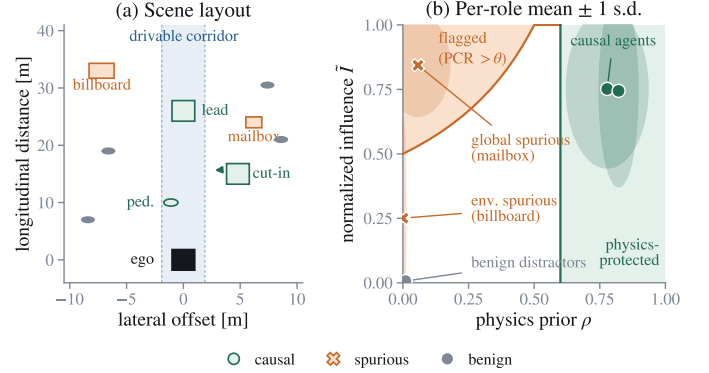


Fig. 4: The SpurGen benchmark. (a) Schematic of a scene in the ego frame: causal agents (lead vehicle, cut-in, pedestrian) occupy the drivable corridor (shaded), while benign distractors and the spurious mailbox/billboard lie off path. (b) Each object category as its mean $\pm$ one standard deviation (marker + ellipse) in the influence–physics plane, over the PCR decision regions: the spurious mailbox falls in the flagged region (high $\tilde{I}$, low ρ), causal agents in the physics-protected region ($\rho \geq \rho_{hi}$), and benign distractors near the origin. No single axis separates all three roles.

distractors have neither, and the spurious mailboxes form the anomalous populations that the audit must isolate: the global mailbox shortcut combines near-zero ρ (mean 0.06) with the highest mean influence of any category by a substantial margin (0.84), whereas the billboard’s influence is diluted across environments (mean 0.25, concentrated in sunny scenes). No single axis separates the mailboxes from both remaining roles, which is the geometric reason that single-signal baselines fail in Table II.

a) *Compared methods.*: A baseline can only enter this comparison if it can be run on a frozen planner, since CADET audits pretrained models without retraining. This rules out the causal-intervention methods of Section II: CausalVAD [7] and Beyond Patterns [9] both retrain the planner (and Beyond Patterns targets prediction, not planning), while generative world-model approaches synthesize scenarios rather than audit a given model’s reliance. A numerical comparison against them would require either retraining a planner for each method or transcribing scores obtained under incomparable protocols, and neither is a valid comparison for a training-free audit. We therefore position them as complementary (Section II) and

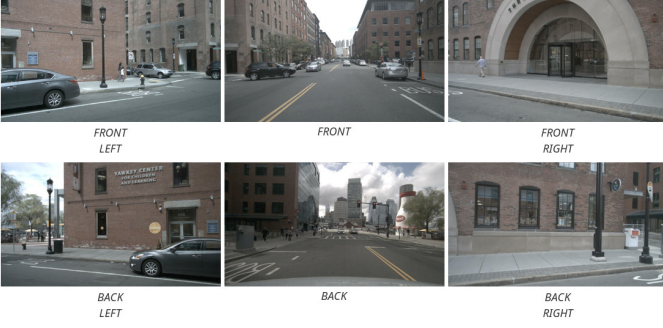


Fig. 5: A nuScenes validation frame: six surround-view camera images that the audited planner consumes. CADET operates on the agent queries the planner detects from these images and on the ego plan it produces.

compare against the training-free signal families on which these methods and CADET are built, each realized as a flagging rule applied to the same frozen planner: (i) **influence-only**, perturbation attribution in the style of occlusion analysis [21] and the effect-vs.-cause diagnosis of de Haan et al. [5], which flags the queries with the largest plan change $\tilde{I}_i > \theta$; (ii) **physics-only**, an ablation of our prior that flags every query the physics deems implausible ($\rho_i < \rho_{lo}$) regardless of model behavior; and (iii) **invariance-only**, an ICP/IRM-style invariance test [8], [20] that performs a per-class one-way ANOVA of raw influence across the three environments and flags high-influence queries of classes for which invariance is rejected ($p < 0.05$). These three baselines instantiate, respectively, the attribution, physics, and invariance families of Table IV, so the comparison isolates the contribution of fusing them under the physics gate. We report flagging precision, recall, and F1 against the construction ground truth.

B. Audit of a Pretrained Planner on nuScenes

a) *Planner and data.*: We audit the public pretrained SparseDrive [3] (stage-2 checkpoint, ResNet-50 backbone, 86M parameters) on the nuScenes-mini validation split (81 keyframes). Each frame provides six surround-view camera images (Fig. 5), from which the planner detects agents and predicts an ego trajectory. The planner is run unmodified, and CADET performs no training and updates no parameters.

b) *Query extraction and ablation.*: SparseDrive exposes vectorized agent queries, and its planning head attends to the top 50 detections by confidence. We audit these agents (those above a confidence of 0.25). The removal operator $f_{\setminus i}$ is realized by zeroing the i -th selected agent’s instance feature inside a cached detection output and re-running only the planning head, while the image backbone, detection, and map outputs are computed once and reused. To keep frames independent and the audit reproducible, the planning head’s temporal queue is reset per frame (single-frame inference). The plan a is the model’s selected ego trajectory, comprising six future waypoints.

c) *Physics prior.*: For each agent we compute ρ_i from its ego-frame box (lateral offset, longitudinal distance, velocity), class, and detection confidence (Section IV-A): an agent that

is off to the side or behind, and thus cannot physically affect the immediate plan, receives $\rho_i \approx 0$, whereas an in-corridor, ahead or closing agent receives a high ρ_i . We use a lane half-width of 2 m and the gate thresholds $\rho_{lo}=0.2$, $\rho_{hi}=0.6$, and $\theta=0.5$.

d) *Metrics.*: Per agent we record the relative plan change r_i under ablation and the score PCR_i . We report the per-agent influence of causal ($\rho \geq \rho_{hi}$) versus spurious ($\rho < \rho_{lo}$) agents and their ratio; the most-influential spurious agent per frame; the Causal Stability Index under ablation of all spurious agents; the number of agents flagged by PCR versus the influence-only baseline; and the open-loop L2 to the ground-truth ego trajectory before and after TCM. Real frames carry no ground-truth irrelevance label, so the decoupled probes of Section IV-B are unavailable here and stability is reported as $CSI-\rho$, which Section VI-B shows to be an upper bound on the true effect. The decoupled probes are used wherever the benchmark supplies ground truth.

e) *Implementation.*: Inference uses the planner’s official code, instrumented to cache the shared backbone, detection, and map computation once and to re-run only the planning head under each query ablation. PCR, the perturbations, and TCM add no gradient computation. Per-frame runtime is dominated by the agent ablations (one planning-head pass each), and the full audit completes in minutes on the single GPU. We release the audit toolkit and the exact configuration.

VI. RESULTS

We first validate the audit mechanism in a controlled synthetic setting where the spurious/causal status of every object is known by construction (Section VI-A). Results on pretrained planners and nuScenes follow (Sections VI-D and VI-E).

A. Method Comparison on SpurGen

Table II compares the four training-free methods of Section V-A on SpurGen (≈ 1790 queries, ≈ 265 spurious), reporting the mean over five random seeds. Each single-signal method fails in the manner predicted by its underlying assumption. The influence-only baseline [21] attains high recall at low precision (0.34 on clean data, rising to only 0.43 at $\sigma=1.5$), because causal agents (including cut-in vehicles approaching the lane) and spurious objects both show high model influence, and perturbation sensitivity alone cannot separate them. The physics-only ablation exhibits the complementary failure mode: it flags every physically implausible object, including the numerous benign distractors on which the planner never relies, and its precision remains below 0.20. The invariance-only test [8], [20] rejects invariance only for environment-dependent mechanisms: it catches the sunny-only billboard shortcut, but it also flags the genuinely causal classes whose behavior is environment-sensitive (rain-cautious cut-ins, night-cautious pedestrians), and it cannot reject invariance for the global mailbox shortcut, so both its precision (0.11) and its recall (0.20) are low. The latter constitutes precisely the faithfulness failure that motivates the external physics anchor. PCR, which requires a query to be both relied upon and physically implausible, attains 0.95 precision at 0.89 recall

TABLE II: Method comparison on SpurGen (400 scenes across three environments; ≈ 1790 queries, ≈ 265 spurious by construction). Precision/recall/F1 are reported for the clean setting, and F1 is then tracked across a perception-noise sweep (σ up to 1.5 on positions, velocities, and confidence). All values are the mean over five random seeds (per-entry standard deviation below 0.04; Fig. 6). Each single-signal method fails as its assumption predicts. Only the fused score attains both high precision and high recall, and its advantage persists under noise.

Method	Clean			F1 under noise σ		
	P	R	F1	0.5	1.0	1.5
influence-only [21]	0.34	0.92	0.50	0.51	0.54	0.58
physics-only (ablation)	0.19	0.91	0.31	0.31	0.31	0.30
invariance-only [20]	0.11	0.20	0.14	0.15	0.16	0.18
PCR (ours)	0.95	0.89	0.92	0.92	0.92	0.90

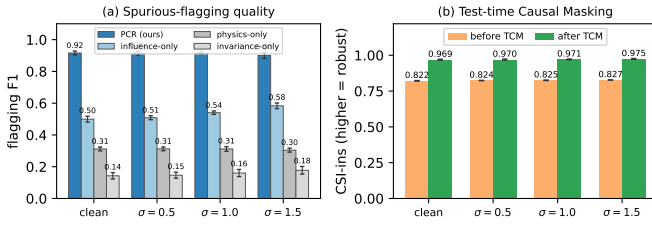


Fig. 6: SpurGen comparison. (a) Flagging F1 of the four training-free methods: influence-only cannot separate causal from spurious reliance, physics-only over-flags benign objects, and invariance-only rejects invariance only for environment-dependent mechanisms while missing the global shortcut. Only the fused PCR score attains both high precision and high recall. (b) Test-time Causal Masking raises stability under the decoupled insertion probe (CSI-ins) from about 0.82 to about 0.97. The residual gap is the 20% of inserted objects the audit does not flag, which are those whose influence falls below θ rather than those the physics prior misjudges. Error bars are the standard deviation over five seeds.

(F1 0.92). Its residual errors are diagnostic: it fails to detect spurious objects on which the planner relies only weakly, as well as those positioned near the lane edge, where ρ is intermediate; and it occasionally mis-flags a slow, distant cut-in that the instantaneous prior cannot yet identify as a hazard. Across the noise sweep, PCR maintains an F1 of 0.90 or above whereas no single-signal baseline exceeds 0.58, confirming that the physics prior constitutes a stable anchor rather than a brittle heuristic. Figure 6(a) visualizes the comparison.

B. Probe Design and the Degeneracy of the Naive Metric

Before reporting the effect of TCM we establish that the metric used to measure it is sound. Table III evaluates the same three modules under the three probes of Section IV-B. Under CSI- ρ , whose target set is the queries with $\rho_i < \rho_{lo}$, the physics-only mask scores exactly 1.00, and it does so on all four planner variants of Table IV. This is an identity rather than a finding: that mask suppresses precisely the set the probe then deletes, so the second deletion changes nothing. The value

TABLE III: Probe design on SpurGen (default planner, mean over five seeds). CSI- ρ selects its targets with the score the modules mask by, so the physics-only mask attains exactly 1.00 by construction. The decoupled probes remove that artifact. Masked % is the share of queries suppressed, which is the cost that a stability score alone does not reveal.

Module	CSI- ρ	CSI-gt	CSI-ins	Masked %
none (base)	0.797	0.795	0.822	—
physics-only	1.000 [†]	0.994	0.993	67
PCR + TCM (ours)	0.983	0.983	0.969	13

[†]degenerate: probe set and masked set coincide.

is therefore uninformative about whether the module removed a spurious dependency. Replacing the probe with the ground-truth set moves the same module to 0.994, and the insertion probe places it at 0.993; the residual gap is the sensitivity that the ρ -defined probe cannot see. The inflation is modest for a selective module such as PCR on the default planner (0.983 against 0.969) and larger where the planner leans harder on the shortcut, reaching 0.064 on the strong-reliance variant. All subsequent stability numbers use CSI-ins.

C. Training-Free Deconfounding (TCM)

On the spurious-containing scenes the planner’s stability under the insertion probe is 0.82 on clean data and 0.83 at $\sigma=1.5$, indicating that a newly introduced physically irrelevant object still displaces the plan appreciably. Applying TCM, which masks the PCR-flagged queries at inference, raises CSI-ins to 0.97 across all settings (Fig. 6(b)). The residual 0.03 is informative: the audit flags 80% of the inserted objects, and the misses are almost entirely queries whose normalized influence falls below θ rather than queries the physics prior scores incorrectly, since objects beyond 3.5 m of lateral offset carry $\rho = 0$ and are still flagged only 81% of the time. The recall bound of the audit is set by the influence threshold, not by the external anchor. Because the hard gate in Eq. (8) seldom flags a physically plausible query, TCM leaves causal agents unaffected and preserves clean-scene behavior, at a cost of one additional forward pass and no parameter updates.

Figure 7 illustrates the effect on the SpurGen scene exhibiting the largest plan correction. Misled by three off-corridor spurious objects, the baseline planner decelerates and advances only 23.0 m over the planning horizon; once TCM masks these objects, it advances 35.9 m along a corridor that is in fact unobstructed, a 12.9 m correction of purely spurious caution. On the pretrained SparseDrive planner (Fig. 8) the same intervention produces a much smaller displacement, because that planner’s spurious reliance is bounded (Section VI-D): TCM removes the flagged reliance while displacing the trajectory only marginally, which is why open-loop L2 fails to register the change. The controlled benchmark exposes the mechanism at full strength, and the real-data audit measures how far a strong contemporary planner is affected.

Table IV casts the comparison as a model $\times$ module matrix: every plug-and-play module is applied, through the identical TCM masking operator, to each planner, and we ask which

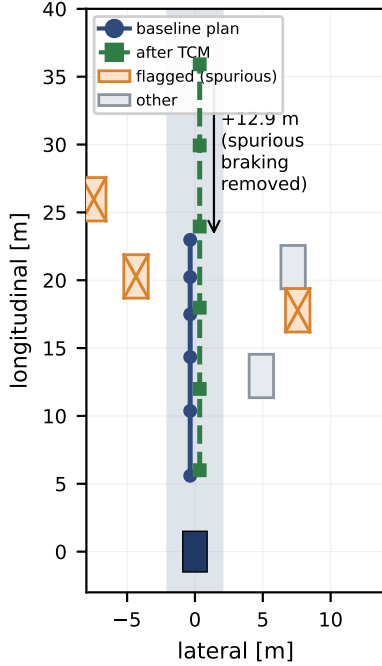


Fig. 7: TCM efficacy on the SpurGen scene with the largest plan change. Relying on three off-corridor spurious objects (orange, crossed), the baseline planner brakes and covers only 23.0m; after TCM masks them it proceeds 35.9m along a corridor that is in fact clear. The gap is the spurious braking TCM removes. Scenes in which the deconfounded plan would reach an in-corridor agent are excluded from this illustration, so the trajectory shown is unobstructed.

module best removes spurious reliance. To probe generality under controlled conditions we use four mock-planner variants spanning weak to strong spurious reliance and a causal-heavy regime (the real SparseDrive planner is audited in Section VI-D); the retraining-based deconfounders and the other test-time methods of Section II cannot be run as a flagging module on a frozen planner, and they are therefore positioned qualitatively in that section rather than benchmarked here. Read under the decoupled probe, the matrix separates the modules by flagging accuracy rather than by stability. CADET attains the highest spurious-flagging F1 on every planner (0.61–0.96, 1.7 to 1.8 times the next-best criterion) while suppressing 8–13% of queries and leaving the causal response at or above its base value. On stability considered alone, the physics-only mask scores higher: it reaches CSI-ins 0.99–1.00 against 0.92–0.97 for CADET, and it too holds CRI at 0.67–0.69. It reaches that stability by suppressing 67% of all queries, of which 72% are benign distractors that the planner never relied upon and only 28% are objects it did rely upon. Its flagging F1 of 0.31 reflects the same thing: the physics prior alone answers which objects cannot matter, not which of them a given planner has come to depend on, and the latter is the audit question. The comparison shows that a stability score cannot be read in isolation, since it is maximized by deleting most of the scene. Occlusion attribution

and ICP/IRM raise stability only by masking indiscriminately, and they degrade CRI to 0.10–0.27 from a baseline of 0.63–0.69, and the count-matched random and confidence controls yield only marginal gains. Taken jointly, CADET is the only criterion that is at once precise ($F1 \geq 0.61$ everywhere), surgical (Masked % ≤ 13) and neutral on causal response, and the physics anchor is what makes this combination transfer: computed from scene geometry, it is unaffected by how a given planner distributes its reliance. The Causal Consistency Score stays ≥ 0.99 throughout and the audit costs 0.025 ms per agent.

D. Audit on a Pretrained Planner (nuScenes)

We apply CADET, without any training, to the public pretrained SparseDrive [3] planner on the nuScenes-mini validation split (81 frames, 4016 audited agent queries). For each detected agent we ablate its instance feature inside the planning head and measure the induced change in the predicted ego trajectory (the relative L2 change r_i), and we score its physics plausibility ρ_i from its position, velocity, class, and detection confidence. Table V summarizes the audit.

Three findings stand out. First, SparseDrive is predominantly causal: a genuinely relevant agent has, on average, $9.5\times$ the per-agent influence of a physically irrelevant one, as expected of a strong planner. Second, spurious reliance is nonetheless real and occasionally severe: the single most-influential physically irrelevant agent in a frame shifts the plan by 0.048 on average, matching the influence of an average causal agent (0.046). The planner therefore does, in specific frames, lean on an agent that physics says cannot matter as heavily as on one that does. Third, consistent with the synthetic study, PCR is more selective than the influence-only baseline (1.63 vs. 2.26 flagged agents per frame, Table VI): the physics hard gate protects high-influence causal agents that the baseline would mislabel as spurious. The audit interface is planner-agnostic: it requires only access to the planner’s agent queries and plan output, so extending the same audit to UniAD [1] and VAD [2] requires only their respective runtime environments, which we leave to an extended study.

Figure 8 illustrates this difference on the validation frames where the two methods disagree most. The influence-only baseline crosses out in-corridor agents whose influence is legitimately high, whereas PCR flags only off-corridor agents the planner should not depend on. Masking those agents (TCM) leaves the plan close to the baseline trajectory, confirming that the removed reliance was not load-bearing for the maneuver.

E. Test-Time Deconfounding and the Blindness of L2 (nuScenes)

Table VI compares two test-time deconfounding variants of the same frozen planner against the unmodified baseline: masking the influence-only-flagged agents, and masking the PCR-flagged agents (TCM). The contrast is informative. TCM raises the Causal Stability Index (0.887 $\rightarrow$ 0.896) while the official open-loop metrics barely move (L2 0.89 $\rightarrow$ 0.96, collision 0.00% $\rightarrow$ 0.08%). The influence-only variant is inferior on every axis: by masking high-influence agents indiscriminately

TABLE IV: Plug-and-play deconfounding as a planner $\times$ module matrix (SpurGen, mean over five seeds). Each module masks agent queries on the same frozen planner through the identical TCM operator and differs only in the flagging criterion; the planners are mock-planner variants with different spurious-reliance profiles. Stability is reported under the decoupled insertion probe (CSI-ins) and, for reference, under the ρ -defined probe (CSI- ρ), on which the physics-only mask is degenerate at 1.00. On every planner CADET attains the best spurious-flagging F1 with the smallest masked fraction and no loss of causal response. Physics-only attains higher stability than CADET on every planner, at the cost of suppressing 67% of all agents; no module dominates on stability alone, which is why Flag-F1 and Masked % must be read alongside it. Occlusion and invariance raise stability only through indiscriminate masking, which degrades CRI. Retraining-based deconfounders (CausalVAD, PLUTO, PlanTF) and other test-time methods (TOAD, Centaur) cannot be run as a flagging module on a frozen planner; they are positioned in Section II.

Planner	Plug-and-play module	Flag-F1	CSI-ins $\uparrow$	CSI- ρ	CRI $\uparrow$	Masked %
weak reliance	none (base)	—	0.91	0.91	0.69	—
	random- k	0.17	0.93	0.92	0.69	8
	confidence- k	0.33	0.94	0.93	0.69	8
	occlusion [21]	0.30	0.97	0.96	0.23	34
	physics-only	0.31	1.00	1.00 [†]	0.69	67
	invariance [8], [20]	0.13	0.94	0.94	0.21	28
	PCR + TCM (ours)	0.61	0.95	0.95	0.69	8
default	none (base)	—	0.82	0.80	0.69	—
	random- k	0.22	0.88	0.87	0.63	13
	confidence- k	0.50	0.92	0.90	0.69	13
	occlusion [21]	0.50	0.97	0.98	0.17	38
	physics-only	0.31	0.99	1.00 [†]	0.69	67
	invariance [8], [20]	0.14	0.84	0.87	0.16	27
	PCR + TCM (ours)	0.92	0.97	0.98	0.69	13
strong reliance	none (base)	—	0.79	0.67	0.66	—
	random- k	0.22	0.83	0.79	0.60	13
	confidence- k	0.52	0.88	0.84	0.67	13
	occlusion [21]	0.55	0.88	0.96	0.10	36
	physics-only	0.31	0.99	1.00 [†]	0.69	67
	invariance [8], [20]	0.15	0.72	0.75	0.14	24
	PCR + TCM (ours)	0.96	0.92	0.99	0.69	13
causal- heavy	none (base)	—	0.86	0.79	0.63	—
	random- k	0.20	0.88	0.85	0.62	10
	confidence- k	0.41	0.92	0.87	0.63	10
	occlusion [21]	0.39	0.90	0.95	0.27	36
	physics-only	0.31	0.99	1.00 [†]	0.67	67
	invariance [8], [20]	0.17	0.84	0.87	0.23	29
	PCR + TCM (ours)	0.75	0.94	0.94	0.64	10

TABLE V: CADET audit of pretrained SparseDrive on nuScenes-mini val (81 frames, 4016 agent queries), inference-only on a single 16 GB GPU. Per-agent influence is the relative plan change when one agent is ablated.

Quantity	Value
Per-agent influence, causal agents ($\rho \geq 0.6$)	0.046
Per-agent influence, spurious agents ($\rho < 0.2$)	0.005
Causal-to-spurious influence ratio	9.46
Most-influential spurious agent / frame	0.048

it removes genuinely causal ones, which lowers CSI (0.881) and yields the largest open-loop degradation (L2 0.99, collision 0.16%). Only the physics-gated mask improves causal robustness, and it does so at a smaller open-loop cost than the baseline, confirming that the gate matters at deployment and not only at audit time.

This is precisely the pathology that motivates our work: open-loop L2 on nuScenes is dominated by ego status [6] and is therefore blind to whether a planner reasons over causal or spurious cues. Both open-loop columns of Table VI

TABLE VI: Test-time deconfounding on nuScenes-mini (SparseDrive, 81 frames). Three variants of the same frozen planner: no audit, masking the influence-only flags, and masking the PCR flags (TCM). L2 and collision are the official SparseDrive open-loop metrics, while CSI is our causal-robustness metric. Real frames carry no ground-truth irrelevance label, so this column uses the ρ -defined probe of Section VI-B and is an upper bound on the true effect; the decoupled probes are available only on SpurGen. TCM is the variant that improves causal robustness (CSI), while the open-loop metrics are nearly invariant to all three interventions.

Variant	Open-loop L2 (m) $\downarrow$				Col. (%) $\downarrow$	CSI $\uparrow$	flag / fr
	1s	2s	3s	Avg			
SparseDrive (base)	0.40	0.84	1.43	0.89	0.00	0.887	0.00
+ Influence-TCM	0.50	0.95	1.53	0.99	0.16	0.881	2.26
+ TCM (ours)	0.46	0.92	1.50	0.96	0.08	0.896	1.63

barely move across the three variants (L2 within 0.1 m, collision within 0.2%), even though their causal behavior differs substantially. CADET’s CSI and per-agent influence

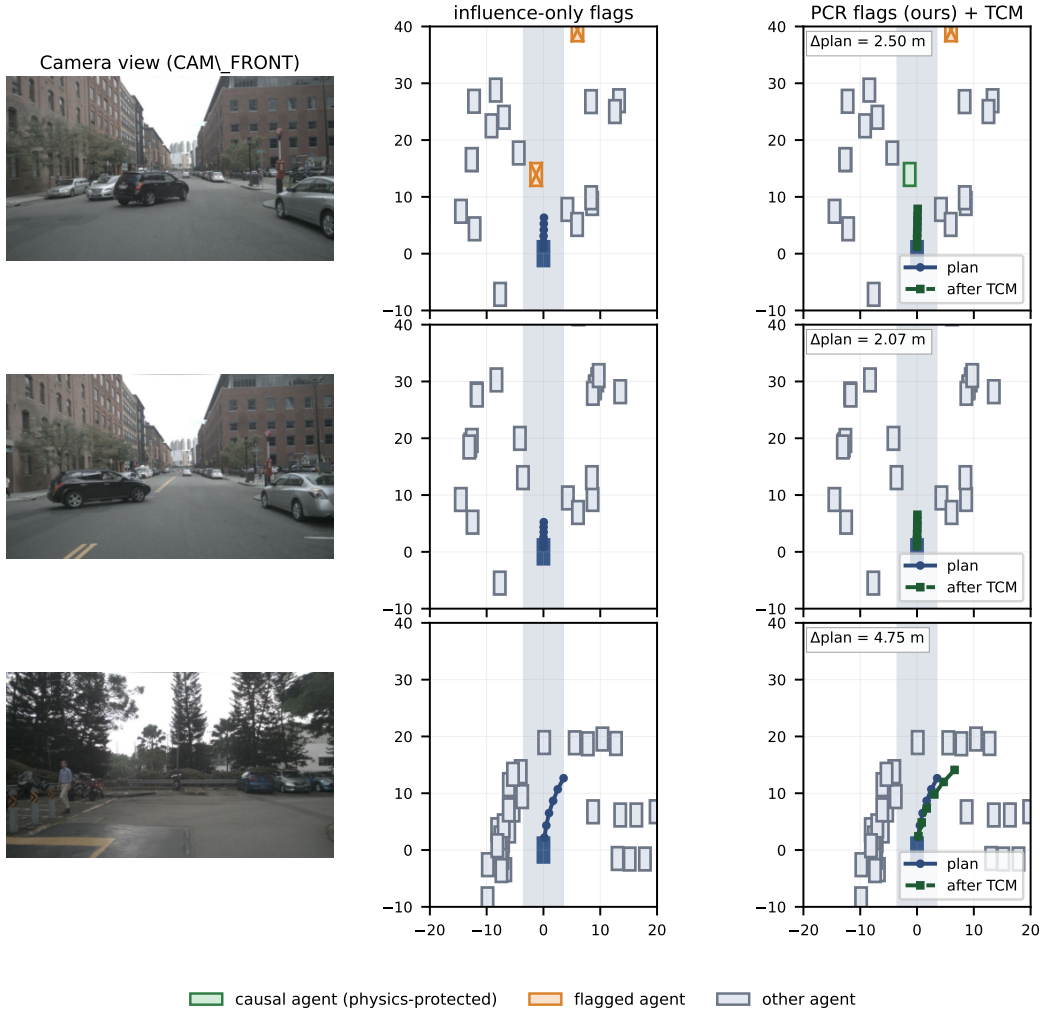


Fig. 8: Agents flagged by each criterion, on the nuScenes frames where the two criteria disagree most. Left: the camera view. Middle: the influence-only baseline flags (crossed boxes) physically plausible, in-corridor agents alongside genuinely spurious ones. Right: PCR flags only agents outside the drivable corridor (orange) while physics-protected causal agents (green) are preserved. The plan after TCM (dashed) stays close to the baseline plan: on this planner the flagged reliance is real but bounded (Section VI-D), so removing it corrects the decision without disturbing the maneuver, which is why open-loop L2 cannot see the change. Figure 7 shows the corresponding plan correction at full strength on the controlled benchmark.

statistics expose a spurious-reliance signal that L2 cannot. This is the empirical case for reporting causal-robustness metrics alongside displacement error. TCM is a diagnostic-driven intervention: it changes the planner’s causal behavior, which open-loop L2 does not measure.

F. What TCM Changes Inside the Planner

To see how TCM reshapes the planner’s behavior, rather than only its output, we measure the planner’s *functional sensitivity*: the raw plan change caused by ablating each agent, before and after masking the flagged set. (SparseDrive uses fused flash-attention, which exposes no softmax map, so a direct attention read-out is unavailable; functional sensitivity measures actual decision impact instead.) Figure 9 splats this sensitivity onto the ground plane for a representative frame: before TCM, the brightest sensitivity sits off the drivable

corridor, on the flagged spurious agents; after TCM it is gone, and only the in-corridor causal reliance remains. The aggregate over the whole split makes the effect precise. Before TCM, the flagged spurious agents are the planner’s most influential inputs (mean influence 0.056 over 150 flagged agents), exceeding even the genuinely causal ones (0.041, $n=64$), a direct internal signature of causal confusion. TCM removes the flagged reliance entirely while leaving causal reliance untouched (0.0407 $\rightarrow$ 0.0409), an order of magnitude above the background spurious region (0.005 $\rightarrow$ 0.004, $n=3151$). The intervention therefore excises the spurious dependency without rerouting the planner’s decisions through other cues or blunting its response to the agents that genuinely matter.

G. Signal-Fusion Ablation

To isolate the contribution of each PCR signal independently of any specific planner, we construct four agent populations

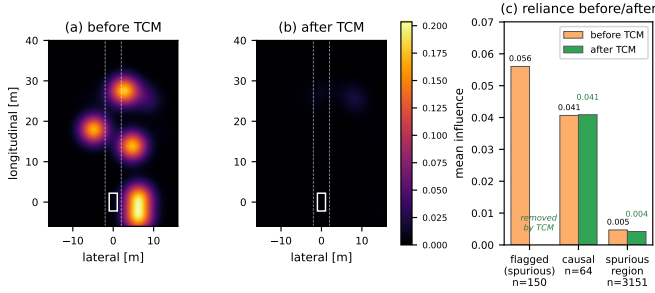


Fig. 9: Internal effect of TCM on the frozen SparseDrive planner. (a,b) Functional-sensitivity heatmap on the ground plane (raw plan change under per-agent ablation) before and after TCM: the off-corridor spurious hot-spots vanish while the in-corridor causal reliance is retained. (c) Mean influence of flagged, causal, and spurious-region agents before versus after TCM. Group sizes n are shown on the axis. Before the intervention the flagged spurious agents are more influential than the causal ones; TCM masks them (so they have no post-TCM bar) while causal influence is left unchanged ($0.041 \rightarrow 0.041$), an order of magnitude above the background spurious region.

TABLE VII: Signal-fusion ablation on 2000 labelled agents (bold = best in column; the shaded row is the full method). The physics prior raises precision by gating causal agents but loses the locally spurious ones; the cross-environment stability term recovers most of them, yielding the best overall F1. The influence-only configuration attains the highest recall only through a lack of selectivity, flagging the majority of queries at a precision of 0.66.

Configuration	Prec.	Rec.	F1	Rec. glob.	Rec. loc.
Influence only	0.660	0.936	0.774	0.942	0.930
+ physics prior	0.988	0.506	0.669	0.818	0.194
+ stability (full PCR)	0.971	0.829	0.894	0.892	0.766

(500 each) with known labels and controlled signal profiles: *causal* (high influence, high ρ , low σ), *global-spurious* (high influence, $\rho \approx 0$, low σ), *local-spurious* (high influence, mid ρ , high σ), and *benign* (low influence). A local-spurious agent is physically plausible-looking, so the physics gate alone does not catch it, but its influence varies across environments. Table VII reports flagging quality as signals are added.

The three rows make each signal’s role explicit. Influence alone flags spurious agents but also causal ones, so its precision is only 0.66. Adding the physics prior raises precision to 0.99 by gating out causal agents, and it catches global-spurious agents (recall 0.82), but it misses local-spurious agents (recall 0.19) because they look physically plausible. Adding the cross-environment stability term recovers most of the local-spurious agents (recall 0.19 $\rightarrow$ 0.77) while keeping precision high (0.97). The same precision effect appears on real data, where the physics gate makes PCR flag fewer agents than the influence-only baseline (1.63 vs. 2.26 per frame, Table VI).

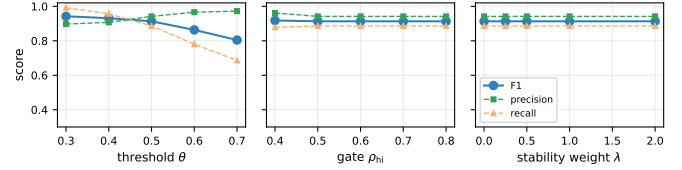


Fig. 10: Parameter sensitivity of PCR on SpurGen. F1 degrades gracefully with the flagging threshold θ and is insensitive to the protection gate ρ_{hi} . The stability weight λ only acts when the cross-environment signal is available (Table VII).

H. Parameter Sensitivity

Figure 10 sweeps the three hyperparameters around their defaults on SpurGen. The flagging threshold θ is the only parameter with a visible effect, and it degrades gracefully: F1 varies from 0.94 at $\theta=0.3$ to 0.81 at $\theta=0.7$, trading precision against recall as expected for a threshold on a calibrated score. The protection gate ρ_{hi} leaves F1 essentially unchanged over $[0.4, 0.8]$, because clearly causal agents receive ρ well above the swept range. The stability weight λ has no effect on SpurGen since the per-query stability signal requires repeated cross-environment estimates of the same query, which the single-frame audit does not provide. Its contribution is isolated in Table VII, where the stability term is active. No parameter requires dataset-specific tuning.

VII. DISCUSSION

A. What the Audit Reveals

Two observations follow from the nuScenes audit. First, causal confusion in a strong contemporary planner is real but bounded: SparseDrive relies on genuinely relevant agents substantially more than on irrelevant ones (a $9.5\times$ per-agent influence ratio), yet in specific frames its single most influential physically irrelevant agent displaces the plan as much as an average causal agent. The failure is therefore concentrated rather than pervasive. An aggregate accuracy metric does not expose this regime, whereas a per-query audit does. Second, because the physics prior is computed from the perception module’s own outputs and not from training-data statistics, CADET localizes this reliance without any labels and without retraining, making it applicable to planners that are already deployed.

B. Open-Loop L2 Does Not Measure Causal Robustness

Applying TCM alters the planner’s causal behavior by removing the flagged spurious reliance, yet the open-loop L2 error with respect to the ground-truth ego trajectory stays unchanged. Because L2 on nuScenes is dominated by ego status [6], it cannot register whether a planner attends to causal or to spurious scene cues. Causal-robustness measures such as the per-agent influence statistics and CSI reported here should therefore accompany displacement error when end-to-end planners are evaluated, and we release them as a benchmark to that end.

C. Scope and Assumptions

CADET audits pretrained planners and, by design, does not retrain them. It is complementary to training-time methods such as CausalVAD [7]. The physics prior is computed from the perception module’s own outputs and is therefore defined over what the upstream detector reports. The perturbations act at the query level, which makes them reproducible and controllable, and they are complementary to the photorealistic counterfactuals produced by a generative world model. The audit criterion is per-query: it flags 80% of the objects introduced by the insertion probe, and the misses are queries whose normalized influence falls below θ even where ρ is zero (Section VI-C). Real frames carry no ground-truth irrelevance label, so the nuScenes audit reports stability under the ρ -defined probe, which is an upper bound; the decoupled probes are supplied by the controlled study (Section VI-B). Future work will extend the audit to more planners, the full nuScenes split, temporal inference, and closed-loop evaluation, and will add a set-level criterion to cover shortcuts that a planner spreads thinly across many queries.

D. Broader Impact

Auditing the causal reliance of a deployed planner supports safer deployment and more honest reporting of robustness. The same machinery could in principle be used to identify exploitable shortcuts in a planner. We mitigate this by framing CADET as a defensive diagnostic and by releasing it alongside the benchmark so that weaknesses are measured and repaired rather than hidden.

E. Reproducibility and Data Use

All experiments are inference-only and run on a single NVIDIA RTX 2000 Ada GPU (16 GB). The synthetic study (SpurGen) depends only on NumPy and scikit-learn, is fully deterministic given a seed, and we report all results as the mean over five seeds. The nuScenes audit uses the public pretrained SparseDrive checkpoint, unmodified, on the public nuScenes-mini split under its non-commercial research license, adding no annotations and training nothing. We fix the gate thresholds ($\rho_{lo}=0.2$, $\rho_{hi}=0.6$, $\theta=0.5$) across all experiments and show in Fig. 10 that no dataset-specific tuning is required. We release the SpurGen generator, the audit and TCM code, the per-experiment configurations, and the scripts that regenerate every table and figure.

VIII. CONCLUSION

We introduced CADET, a training-free framework that measures spurious-correlation reliance in pretrained end-to-end driving planners and localizes it to individual planning queries. Its central idea is to use a physics-geometric prior, computed from the perception module’s own outputs, as an external anchor that is independent of the training distribution and therefore resists the global spurious correlations on which observational signals fail. From this prior we derived a per-query audit score (PCR), a counterfactual robustness benchmark (CSI, CRI, CCS), and a test-time deconfounding

operation (TCM), all of which require only forward passes and run on a single 16 GB GPU.

In a controlled study where the status of every object is fixed by construction, PCR separates spurious from causal reliance at a precision of 0.95 against 0.34 for an influence-only baseline, and it is the only training-free criterion that reaches this precision while suppressing under 13% of queries and leaving causal response intact. Stability alone does not distinguish the criteria, since an indiscriminate physics mask attains a higher stability score by suppressing 67% of all queries, of which 72% are objects the planner never relied upon. What separates the criteria is how accurately they identify the queries a given planner has in fact come to depend on, which is the audit question. On nuScenes, CADET shows that a strong pretrained planner is predominantly causal yet exhibits measurable and occasionally severe spurious reliance, and that suppressing the flagged agents leaves the open-loop L2 error unchanged, which confirms that displacement error is insensitive to causal robustness.

Future work will use closed-loop evaluation to measure the behavioral cost of the dependencies the audit locates, which open-loop displacement cannot register, and will extend the audit to additional planners, the full nuScenes split, and temporal settings. Combining query-level perturbations with generative counterfactuals, and coupling the audit with training-time deconfounding so that flagged spurious dependencies inform retraining, are further directions. We release the code, the benchmark, and the audit toolkit to support them.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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